Dataset Strategy & Initial Data Exploration

Brain CT Hemorrhage Segmentation with Active Learning

1. Project Objective

The objective of this project is to develop a label-efficient active learning pipeline for brain CT hemorrhage segmentation. The model will be trained using a labelled segmentation dataset and later used to identify informative cases from a separate dataset through uncertainty scoring. The experiment will focus on the five target hemorrhage classes: IPH, SAH, EDH, SDH and IVH, along with background.

2. Proposed Dataset Strategy

Dataset	Role	Reason for Selection
BHSD	Primary training dataset	Provides 3D brain CT data with voxel-level hemorrhage annotations and is suitable for multi-class segmentation.
RSNA 3D / INSTANCE 2022	Separate AL pool	Provides an independent source of CT studies for uncertainty scoring and evaluation. Final choice will be confirmed after checking label format, accessibility and compatibility.

3. Dataset Selection Rationale

BHSD is selected as the main training dataset because the project requires pixel or voxel-level masks for segmentation. Its multi-class hemorrhage annotations make it the most direct fit for the proposed task.

For the active learning pool, RSNA 3D and INSTANCE 2022 are being considered as the independent dataset source. The pool must remain completely separate from training. The final pool will be fixed after checking the available annotations, dataset format, and compatibility with the BHSD data.

4. Data Separation Plan

The training and active learning datasets will be strictly separated. The model will not be trained on the AL pool before uncertainty scoring. Study identifiers will be checked to ensure there is no accidental overlap between the datasets.

5. Initial Data Exploration

We explored different brain CT datasets like BHSD, RSNA, PhysioNet, CQ500, and INSTANCE 2022 and compared their data and labels. We also went through the five types of brain hemorrhage: IPH, SAH, EDH, SDH, and IVH and their appearance in CT scans. Along with this, we loaded and checked some DICOM images.